

CMPT276 Group 20, Damiano and Jeff

Rob the Block: HW 4 Report

1. We found that there were duplicated code when it comes to movement with our enemies, dogs and homeowners. We a condition is met for both the homeowner and dog they will “chase” to a location on the map. Both of these classes had the same path finding algorithm which was copied over. We took the algorithm from Dog.java and Homeowner.java and put them into its own class ChaseAlgorithm.java. This makes the code cleaning with less duplication
2. The “Game.java” had become a god class that has everything in it. In this case, we decided to take out the camera and give it its own class, GameCamera.java. Along with this the cameras methods and code weren’t all in one place, making it harder to understand where and how it worked. Now with GameCamera.java, the camera moves with the character, making our camera movement easier to understand
3. Within Game.java there was a method render() which had multiple responsibilities and being very long. Within this it drew all of the sprites, but also Game.java also had some classes that was also drawing sprites which created a bit of spaghetti code since there were classes in multiple places doing the same thing. We were able to take the logic from Game.java and render() and put them into a new class called GameDraw.java, which draws all the sprites.
4. Another case of duplicate code was found when it came to loading out levels. In our classes restartLevel(), nextLevel() and startGame() all had the same 4 lines in it which were just things to start the level. Taking these lines out reduced duplicated code within these classes, and they were put in a new method called loadLevel() which loads each required thing in the game.
5. In Game.java, we found a data clump of textures all being initialized. We abstracted all the textures into their own separate class (TextureRenderer.java), using getters where needed to display the textures.
6. We found a class called DetectionPunishment.java within our objects folder that was used zero times throughout the project. The class was promptly removed from the project.
7. Found a duplicate method within Game.java and burglar.java called isOutofCoins() which checks if the total score of the burglar is greater than 0. Removed the Game.java version of the method as score is a parameter of burglar.

8. The InputKeys class contained two instance variables, X and Y, that were only ever used within handleInput(), making them unnecessary fields of the class. These were replaced with local variables x and y declared directly inside the if (moved) block where they are needed. Additionally, two separate if (moved) blocks that handled reward collection and dog toy interaction were merged into a single block, eliminating the duplicated code.